



## POLYTECHNIC UNIVERSITY OF THE PHILIPPINES

### Data Dictionary

This data dictionary presents the information stored in the CrayCare Cloud Firestore database. Each table identifies the field name, data type, description, and required or default value. It explains in simple terms how the mobile application, backend services, and ESP32 hardware use each field.

The database uses common data types such as string, number, boolean, timestamp, array, map, and null. The table names are written in a readable form, while the field names remain exactly as they appear in the system.

**Table: Users**

| Field Name | Data Type     | Description                                                                                            | Constraints / Default        |
|------------|---------------|--------------------------------------------------------------------------------------------------------|------------------------------|
| full_name  | string        | User's complete display name.                                                                          | Required                     |
| email      | string        | Email address used for authentication and account recovery.                                            | Required; unique             |
| role       | string        | Access role. Only one active administrator account is intended; owners operate assigned tank features. | owner   admin                |
| status     | string        | Controls whether the account may use protected app features.                                           | active   disabled            |
| tank_id    | string        | Tank owned by the user; normally equal to the owner's UID.                                             | Owner: required; Admin: null |
| photo_url  | string        | Main link to the user's profile image. photoUrl is kept only so older app versions can still read it.  | Optional                     |
| fcmTokens  | array<string> | Notification token for each signed-in device.                                                          | Default: empty               |
| fcmToken   | string        | Single-token compatibility field for older app versions.                                               | Legacy; optional             |
| created_at | timestamp     | Time the profile document was created.                                                                 | Server timestamp             |

**Table: Notification Settings**

| Field Name  | Data Type | Description                                                      | Constraints / Default |
|-------------|-----------|------------------------------------------------------------------|-----------------------|
| sound       | boolean   | Enables notification sound where permitted by the device.        | Default: true         |
| vibration   | boolean   | Enables vibration where permitted by the device.                 | Default: true         |
| critical    | boolean   | Enables critical water-quality notifications.                    | Default: true         |
| warning     | boolean   | Enables warning-level water-quality notifications.               | Default: true         |
| feeding     | boolean   | Enables feeding reminders and completed/skipped feeding results. | Default: true         |
| sampling    | boolean   | Enables tank sampling reminders.                                 | Default: true         |
| operational | boolean   | Enables actuator, device, and sensor-recovery updates.           | Default: true         |
| updated_at  | timestamp | Time the preferences were last updated.                          | Server timestamp      |



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**Table: Notification Markers**

| Field Name | Data Type                   | Description                                            | Constraints / Default |
|------------|-----------------------------|--------------------------------------------------------|-----------------------|
| markerKey  | string                      | Stable identifier for the reminder or scheduled check. | Required              |
| value      | boolean<br>string<br>number | Marker state or last processed value.                  | Required              |
| updated_at | timestamp                   | Time the marker was last changed.                      | Server timestamp      |

**Table: Current Hardware Owner**

| Field Name  | Data Type     | Description                                                           | Constraints / Default     |
|-------------|---------------|-----------------------------------------------------------------------|---------------------------|
| uid         | string   null | UID of the currently assigned owner.                                  | Active owner UID or null  |
| tank_id     | string   null | Tank receiving data and device operations from the assigned hardware. | Must match user's tank_id |
| assigned_by | string        | Administrator who made the assignment.                                | Admin UID                 |
| assigned_at | timestamp     | Time the assignment became effective.                                 | Server timestamp          |

**Note:** When no owner is assigned, both uid and tank\_id are null. If the assigned owner is disabled, both values are cleared together.

**Table: Tanks**

| Field Name                  | Data Type     | Description                                                      | Constraints / Default |
|-----------------------------|---------------|------------------------------------------------------------------|-----------------------|
| owner_uid                   | string        | UID of the owner responsible for the tank.                       | Required              |
| current_batch_id            | string   null | Identifier of the active production batch.                       | Optional              |
| stocking_date               | timestamp     | Date and time the current grow-out cycle was stocked.            | Optional              |
| last_sample_date            | timestamp     | Date of the latest recorded sample.                              | Optional              |
| sample_count                | number        | Number of samples recorded for the current cycle.                | Default: 0            |
| initial_population          | number        | Counted population stocked at the beginning of the cycle.        | Non-negative integer  |
| initial_total_sample_weight | number        | Total weight of the baseline sample.                             | Optional; grams       |
| initial_total_sample_length | number        | Total length of the baseline sample.                             | Optional; centimeters |
| is_initialized              | boolean       | Whether tank and baseline production data have been initialized. | Default: false        |
| created_at                  | timestamp     | Time the tank document was created.                              | Server timestamp      |



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**Table: Latest Sensor Readings**

| Field Name           | Data Type | Description                                                                                          | Constraints / Default |
|----------------------|-----------|------------------------------------------------------------------------------------------------------|-----------------------|
| temperature          | number    | Current water temperature.                                                                           | degrees Celsius       |
| ph_level             | number    | Current acidity or alkalinity reading.                                                               | pH                    |
| dissolved_oxygen     | number    | Current dissolved oxygen concentration.                                                              | mg/L                  |
| turbidity            | number    | Current water turbidity.                                                                             | NTU                   |
| turbidity_air        | number    | Air/reference reading used by the turbidity integration.                                             | Optional              |
| water_level          | number    | Current pond water level.                                                                            | centimeters           |
| feed_level           | number    | Estimated percentage of feed remaining in the hopper; operational only and excluded from WQA inputs. | 0-100 percent         |
| estimated_feed_grams | number    | Estimated remaining feed calculated from feed level and calibrated hopper capacity.                  | Optional; grams       |
| buffered_entries     | number    | Number of locally buffered readings reported by the ESP32.                                           | Optional              |
| recorded_at          | timestamp | Time when the ESP32 collected the sensor reading.                                                    | Required              |

**Table: Daily Sensor Summaries**

| Field Name                       | Data Type     | Description                                                                             | Constraints / Default |
|----------------------------------|---------------|-----------------------------------------------------------------------------------------|-----------------------|
| date_key                         | string        | Calendar key represented by this daily document.                                        | YYYY-MM-DD            |
| summary_version                  | number        | Schema/version number of the summary computation.                                       | Current: 1            |
| summary_sanitized                | boolean       | Shows that invalid sensor values have already been removed from the summary.            | Required              |
| summary_complete                 | boolean       | Indicates whether the daily summary is ready for optimized reads.                       | Required              |
| sample_count                     | number        | Number of valid history entries included.                                               | Default: 0            |
| processed_entry_ids              | array<string> | IDs of readings already included, preventing the same reading from being counted twice. | Default: empty        |
| temp_min/max/avg/sum/count       | number fields | Daily temperature statistics.                                                           | When data exists      |
| pH_min/max/avg/sum/count         | number fields | Daily pH statistics.                                                                    | When data exists      |
| DO_min/max/avg/sum/count         | number fields | Daily dissolved-oxygen statistics.                                                      | When data exists      |
| turbidity_min/max/avg/sum/count  | number fields | Daily turbidity statistics.                                                             | When data exists      |
| waterLevel_min/max/avg/sum/count | number fields | Daily water-level statistics.                                                           | When data exists      |
| updated_at                       | timestamp     | Time the daily summary was last maintained.                                             | Server timestamp      |



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**Table: Sensor History Entries**

| Field Name                         | Data Type | Description                                               | Constraints / Default |
|------------------------------------|-----------|-----------------------------------------------------------|-----------------------|
| temp_min,<br>temp_max,<br>temp_avg | number    | Temperature statistics for the ten-minute window.         | degrees Celsius       |
| pH_min, pH_max,<br>pH_avg          | number    | pH statistics for the ten-minute window.                  | pH                    |
| DO_min, DO_max,<br>DO_avg          | number    | Dissolved-oxygen statistics for the window.               | mg/L                  |
| turbidity_min/max/avg              | number    | Turbidity statistics for the window.                      | NTU                   |
| waterLevel_min/max/avg             | number    | Water-level statistics for the window.                    | centimeters           |
| feed_level                         | number    | Feed level reading; excluded from WQA model features.     | Optional; percent     |
| estimated_feed_grams               | number    | Estimated feed remaining during the aggregate window.     | Optional; grams       |
| recorded_at                        | timestamp | Original capture time of the completed ten-minute window. | Required              |

**Table: Sensor Thresholds**

| Field Name            | Data Type | Description                                                                             | Constraints / Default |
|-----------------------|-----------|-----------------------------------------------------------------------------------------|-----------------------|
| min_value             | number    | Lower boundary of the configured normal/ideal range.                                    | Required              |
| max_value             | number    | Upper boundary of the configured normal/ideal range.                                    | Required              |
| critical_value        | number    | Percentage below which the feed level is critical. This alone does not prevent feeding. | feed_level only       |
| hopper_capacity_grams | number    | Calibrated feed mass when the hopper is full; used to estimate remaining grams.         | feed_level only       |
| updated_at            | timestamp | Time the threshold configuration was last updated.                                      | Server timestamp      |

**Note:** The feed level statuses are Normal (>20%), Low (11-20%), Critical (1-10%), and Empty (0%). Feeding is skipped only when feed is empty, unavailable, or estimated grams are below the requested amount.

**Table: Actuators**

| Field Name    | Data Type | Description                                 | Constraints / Default                         |
|---------------|-----------|---------------------------------------------|-----------------------------------------------|
| control_mode  | string    | Operating mode requested by the owner.      | on   off   auto                               |
| current_state | string    | Actual relay state reported by the ESP32.   | on   off                                      |
| last_changed  | number    | Time the device state or mode last changed. | milliseconds since January 1, 1970; 0 = never |



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**Table: Actuator Logs**

| Field Name    | Data Type           | Description                                                             | Constraints / Default                          |
|---------------|---------------------|-------------------------------------------------------------------------|------------------------------------------------|
| actuator_type | string              | Device associated with the event.                                       | pump   aerator1   aerator2                     |
| action        | string              | Human-readable description of the operation.                            | Required                                       |
| type          | string              | Machine-readable operation type.                                        | on   off   auto                                |
| logged_at     | number<br>timestamp | Time the operation occurred; older timestamp formats can still be read. | New writes: milliseconds since January 1, 1970 |

**Table: Feeder Status**

| Field Name           | Data Type           | Description                                                    | Constraints / Default                                                                |
|----------------------|---------------------|----------------------------------------------------------------|--------------------------------------------------------------------------------------|
| status               | string              | Current or most recent feeder state.                           | idle   checking_feed_level   dispensing   completed   skipped_insufficient   blocked |
| status_reason        | string              | Human-readable explanation when feeding is blocked or skipped. | Optional                                                                             |
| command_id           | string              | Manual command correlated with the current outcome.            | Optional                                                                             |
| dispenseCount        | number              | Cumulative completed dispensing cycles.                        | Default: 0                                                                           |
| lastSeen             | number              | Most recent feeder heartbeat/status time.                      | milliseconds since January 1, 1970                                                   |
| last_dispensed_at    | number<br>timestamp | Time of the last completed dispensing cycle.                   | Optional                                                                             |
| last_dispensed_grams | number              | Estimated amount associated with the last completed cycle.     | Optional; estimated grams                                                            |
| feed_level           | number              | Latest hopper level available to the feeder.                   | 0-100 percent                                                                        |
| estimated_feed_grams | number              | Estimated mass remaining in the hopper.                        | Optional; grams                                                                      |

**Table: Feeder Schedules**

| Field Name | Data Type                   | Description                                                                       | Constraints / Default   |
|------------|-----------------------------|-----------------------------------------------------------------------------------|-------------------------|
| time       | string                      | Human-readable scheduled time.                                                    | H:MM                    |
| ampm       | string                      | Meridiem indicator.                                                               | AM   PM                 |
| timeValue  | number                      | Minutes since midnight, used for ordering and overlap validation.                 | 0-1439                  |
| grams      | number   null               | Estimated feed amount. Null uses the default 20 g.                                | 20-200; multiples of 20 |
| days       | array<string>   string mask | Recurring days of the week. Firmware may use a Sunday-first seven-character mask. | At least one active day |



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| Field Name         | Data Type | Description                                                                                                       | Constraints / Default                               |
|--------------------|-----------|-------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------|
| enabled            | boolean   | Whether the schedule is active.                                                                                   | Default: true                                       |
| isDone             | boolean   | Older field that becomes true only when the latest feeding was completed. Use last_outcome for the latest result. | For older app versions                              |
| created_at         | timestamp | Time the schedule was created.                                                                                    | Server timestamp                                    |
| effective_at_ms    | number    | Time the schedule was created, edited, or enabled again. Earlier schedule times will not be marked as missed.     | UTC time stored as milliseconds                     |
| last_outcome       | string    | Latest confirmed result of the schedule.                                                                          | completed   skipped_insufficient   blocked   failed |
| last_occurrence_at | number    | Scheduled minute associated with last_outcome.                                                                    | UTC time stored as milliseconds                     |

**Table: Feeding History**

| Field Name                | Data Type          | Description                                                                                      | Constraints / Default                               |
|---------------------------|--------------------|--------------------------------------------------------------------------------------------------|-----------------------------------------------------|
| action                    | string             | Human-readable feeding-event description.                                                        | Required                                            |
| type                      | string             | Source/category of the feeder event.                                                             | auto   manual   missed   error                      |
| status                    | string             | Terminal result reported by the ESP32.                                                           | completed   skipped_insufficient   blocked   failed |
| command_id                | string             | Originating manual command ID.                                                                   | Optional                                            |
| schedule_key              | string             | Originating schedule document ID.                                                                | Scheduled feeds only                                |
| schedule_time             | string             | Scheduled feeding time saved in the history record.                                              | Scheduled feeds only                                |
| occurrence_at             | number             | Original scheduled time, even if the device sends the result later after reconnecting.           | UTC time stored as milliseconds                     |
| requested_grams           | number             | Feed amount requested by the command or schedule.                                                | Required for execution                              |
| estimated_dispensed_grams | number             | Estimated amount based on the feeder motor cycle. It is not an exact weight measured by a scale. | Completed only                                      |
| amount_basis              | string             | Shows that the dispensed amount was estimated from the feeder motor cycle.                       | servo_cycle_estimate                                |
| estimated_available_grams | number             | Estimated hopper mass available before execution.                                                | Optional                                            |
| feed_level_before         | number             | Hopper percentage before dispensing.                                                             | Optional; percent                                   |
| feed_level_after          | number             | Hopper percentage after dispensing.                                                              | Optional; percent                                   |
| level_change_detected     | boolean            | Supporting indication of feed movement; not proof of exact grams.                                | Optional                                            |
| verification_note         | string             | Additional interpretation of the before/after feed-level check.                                  | Optional                                            |
| logged_at                 | number   timestamp | Original event time. Completed logs alone contribute to Consumption Today.                       | New writes: milliseconds since January 1, 1970      |



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**Table: Feeder Commands**

| Field Name   | Data Type     | Description                                                                    | Constraints / Default   |
|--------------|---------------|--------------------------------------------------------------------------------|-------------------------|
| command_type | string        | Requested manual action.                                                       | feed_now                |
| grams        | number   null | Requested amount; null uses the default 20 g.                                  | 20-200; multiples of 20 |
| issued_by    | string        | User who created the request.                                                  | Owner UID               |
| issued_at    | timestamp     | Official time when the feeding request was created.                            | Server timestamp        |
| expires_at   | timestamp     | Application deadline; firmware also rejects requests more than 60 seconds old. | Required for new writes |

**Note:** Deletion of a command is not proof of completed feeding. The UI confirms outcomes using a matching command\_id in feeder status/log records.

**Table: Feeder Notification Records**

| Field Name              | Data Type | Description                                                                      | Constraints / Default |
|-------------------------|-----------|----------------------------------------------------------------------------------|-----------------------|
| uid                     | string    | Recipient owner UID.                                                             | Required              |
| push_attempt_claimed_at | timestamp | Time when the backend reserved this feeding result for one notification attempt. | Server timestamp      |

**Note:** This record confirms that the backend attempted the push notification. It does not guarantee that the phone received it. The notification is still saved in the app inbox.

**Table: Water Quality Assessments**

| Field Name      | Data Type | Description                                                                                                  | Constraints / Default                             |
|-----------------|-----------|--------------------------------------------------------------------------------------------------------------|---------------------------------------------------|
| uid             | string    | Owner UID associated with the assessment.                                                                    | Required                                          |
| tank_id         | string    | Tank evaluated by the assessment.                                                                            | Required                                          |
| level           | string    | Final user-facing assessment after safety and data-availability handling.                                    | Good   Moderate   Poor   Critical   Insufficient  |
| model_level     | string    | Water-quality class produced by the Random Forest model.                                                     | Good   Moderate   Poor   Critical                 |
| rule_level      | string    | Optional rule result used as an additional safety check when recent readings show risk.                      | Optional                                          |
| safety_override | boolean   | True when the safety floor raises the displayed severity.                                                    | Default: false                                    |
| source          | string    | Shows whether the result came from the trained model, a backup rule, insufficient readings, or old readings. | Required                                          |
| model_algorithm | string    | Name of the method used to produce the assessment.                                                           | RandomForestClassifier   Rule-based   Not applied |



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| Field Name              | Data Type     | Description                                                                           | Constraints / Default                             |
|-------------------------|---------------|---------------------------------------------------------------------------------------|---------------------------------------------------|
| model_version           | string        | Version embedded in the deployed model artifact.                                      | Required when model used                          |
| training_data_origin    | string        | Origin of model-training records, such as synthetic prototype or actual pond history. | Required                                          |
| label_origin            | string        | Origin of training labels; documents current prototype limitations.                   | Required                                          |
| model_feature_count     | number        | Number of engineered model inputs.                                                    | Required when model used                          |
| analysis_window_minutes | number        | Historical duration analyzed for the assessment.                                      | Default: 60                                       |
| data_status             | string        | Shows whether enough recent sensor history was available for the assessment.          | ready   insufficient   stale                      |
| source_recorded_at      | string   null | Timestamp of the newest actual source reading.                                        | ISO-8601                                          |
| source_age_seconds      | number        | Age of the newest source reading at processing time.                                  | Non-negative                                      |
| confidence              | number        | Model confidence; omitted or not presented as ML confidence for fallback results.     | Model output only                                 |
| driver                  | string        | Primary parameter driving the assessment.                                             | DO   pH   temp   turbidity   waterLevel   overall |
| driver_label            | string        | Human-readable driver name.                                                           | Required                                          |
| driver_value            | number        | Representative value of the primary driver.                                           | Optional                                          |
| driver_unit             | string        | Measurement unit of driver_value.                                                     | Optional                                          |
| driver_min              | number        | Minimum driver value in the analyzed window.                                          | Optional                                          |
| driver_max              | number        | Maximum driver value in the analyzed window.                                          | Optional                                          |
| problem                 | string        | Short statement of the detected water-quality concern.                                | Optional                                          |
| insight                 | string        | Trend-aware explanation for the user.                                                 | Optional                                          |
| action                  | string        | Recommended monitoring or management response.                                        | Optional                                          |
| concerns                | array   map   | Primary parameter concerns identified during interpretation.                          | Optional                                          |
| secondary_concerns      | array   map   | Additional lower-priority concerns.                                                   | Optional                                          |
| ts_epoch                | number        | Numeric processing time used to arrange assessment records in order.                  | UTC time stored as a number                       |
| timestamp               | string        | Human-readable assessment processing time.                                            | ISO-8601                                          |

**Note:** The assessment needs at least six valid ten-minute records in a row. If the newest reading is more than 20 minutes old, the result becomes Insufficient or Stale. Feed level is not included in the Water Quality Assessment.





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**Table: Production Batches**

| Field Name           | Data Type | Description                                                                               | Constraints / Default           |
|----------------------|-----------|-------------------------------------------------------------------------------------------|---------------------------------|
| batch_status         | string    | Current lifecycle state of the batch.                                                     | active   harvested   superseded |
| stocking_date        | timestamp | Stocking date; the setup calendar day is Day 0.                                           | Required                        |
| harvest_date         | timestamp | Date the batch was harvested.                                                             | Optional                        |
| initial_count        | number    | Counted population stocked at Day 0.                                                      | Non-negative integer            |
| current_count        | number    | Counted population after recorded mortality and harvest events.                           | Non-negative integer            |
| harvest_count        | number    | Cumulative crayfish harvested.                                                            | Default: 0                      |
| total_mortality      | number    | Cumulative recorded mortality.                                                            | Default: 0                      |
| harvest_weight_grams | number    | Cumulative recorded harvest weight in grams.                                              | Optional                        |
| initial_abw          | number    | Average body weight from the baseline sample.                                             | Optional; grams                 |
| initial_abl          | number    | Average body length from the baseline sample.                                             | Optional; centimeters           |
| final_abw            | number    | Final average body weight at harvest.                                                     | Optional; grams                 |
| final_abl            | number    | Final average body length at harvest.                                                     | Optional; centimeters           |
| days_in_culture      | number    | Calendar days since stocking, updated at midnight rather than as rolling 24-hour periods. | Default: 0                      |
| sample_count         | number    | Number of non-baseline sampling records.                                                  | Default: 0                      |
| initial_total_weight | number    | Total weight of the baseline sample.                                                      | Optional; grams                 |
| initial_total_length | number    | Total length of the baseline sample.                                                      | Optional; centimeters           |
| created_at           | timestamp | Time the batch record was created.                                                        | Server timestamp                |

**Table: Sampling Records**

| Field Name      | Data Type | Description                                                                                | Constraints / Default |
|-----------------|-----------|--------------------------------------------------------------------------------------------|-----------------------|
| sampling_date   | timestamp | Date the sample was collected.                                                             | Required              |
| avg_body_weight | number    | Average sample weight: total_weight divided by sample_size.                                | grams                 |
| avg_body_length | number    | Average sample length: total_length divided by sample_size.                                | centimeters           |
| sample_size     | number    | Number of crayfish physically measured in the sample.                                      | Positive integer      |
| total_weight    | number    | Combined weight of sampled crayfish.                                                       | grams                 |
| total_length    | number    | Combined measured length of sampled crayfish.                                              | centimeters           |
| biomass         | number    | Estimated tank biomass calculated from current counted population and average body weight. | Estimated grams       |
| live_count      | number    | Current counted live population used in the biomass estimate.                              | Counted population    |
| is_baseline     | boolean   | True for the Day 0 baseline; false for weekly samples.                                     | Default: false        |



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| Field Name | Data Type | Description                           | Constraints / Default |
|------------|-----------|---------------------------------------|-----------------------|
| created_at | timestamp | Time the sampling record was created. | Server timestamp      |

**Note:** Weekly sampling begins seven calendar days after stocking and continues every seven days from the previous weekly sample. The baseline record is excluded from Week N counting.

**Table: Mortality Records**

| Field Name      | Data Type | Description                                         | Constraints / Default |
|-----------------|-----------|-----------------------------------------------------|-----------------------|
| mortality_date  | timestamp | Date the dead crayfish were discovered or recorded. | Required              |
| mortality_count | number    | Number of mortalities in the event.                 | Positive integer      |
| created_at      | timestamp | Time the record was created.                        | Server timestamp      |

**Table: Harvest Records**

| Field Name      | Data Type | Description                                     | Constraints / Default |
|-----------------|-----------|-------------------------------------------------|-----------------------|
| batch_id        | string    | Parent batch identifier retained in the record. | Required              |
| harvest_date    | timestamp | Date the harvest was performed.                 | Required              |
| harvest_count   | number    | Number of crayfish harvested.                   | Positive integer      |
| total_weight_kg | number    | Combined harvest weight.                        | kilograms             |
| abw_grams       | number    | Average harvested body weight.                  | grams                 |
| created_at      | timestamp | Time the record was created.                    | Server timestamp      |

**Table: Notifications**

| Field Name | Data Type | Description                                                    | Constraints / Default                                 |
|------------|-----------|----------------------------------------------------------------|-------------------------------------------------------|
| uid        | string    | UID of the intended recipient.                                 | Required                                              |
| notif_type | string    | Notification category used for grouping and preference checks. | critical   warning   reminder   device_auto   general |
| title      | string    | Short notification heading.                                    | Required                                              |
| body       | string    | Detailed notification message.                                 | Required                                              |
| is_read    | boolean   | Whether the user opened or marked the notification as read.    | Default: false                                        |
| created_at | timestamp | Time the notification record was created.                      | Server timestamp                                      |

**Note:** Completed and skipped-insufficient feeder logs create inbox records and immediate FCM pushes when the user's Feeding preference is enabled.



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**Table: Current ESP32 Sensor Data**

| Field Name              | Data Type | Description                                                                   | Constraints / Default              |
|-------------------------|-----------|-------------------------------------------------------------------------------|------------------------------------|
| hardwareId              | string    | Stable identifier of the reporting hardware.                                  | Required                           |
| source_tank_id          | string    | Tank bound to the hardware when the reading was captured.                     | Required                           |
| source_owner_uid        | string    | Owner bound to the hardware at capture time.                                  | Required                           |
| source_assignment_at_ms | number    | Owner and tank assignment used to reject data from an old or different owner. | milliseconds since January 1, 1970 |
| captured_at_ms          | number    | Original capture time when the device clock is trusted.                       | milliseconds since January 1, 1970 |
| temperature             | number    | Live temperature value.                                                       | Optional per validity              |
| ph_level                | number    | Live pH value.                                                                | Optional per validity              |
| dissolved_oxygen        | number    | Live dissolved oxygen value.                                                  | Optional per validity              |
| turbidity               | number    | Live water turbidity value.                                                   | Optional per validity              |
| turbidity_air           | number    | Turbidity reference/air value.                                                | Optional                           |
| water_level             | number    | Live pond water-level value.                                                  | Optional per validity              |
| feed_level              | number    | Live hopper percentage.                                                       | Optional per validity              |
| estimated_feed_grams    | number    | Estimated hopper mass.                                                        | Optional                           |
| buffered_entries        | number    | Count of device readings waiting in the local buffer.                         | Default: 0                         |

**Table: ESP32 Sensor Data History**

| Field Name                | Data Type | Description                                                                                                                                    | Constraints / Default              |
|---------------------------|-----------|------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|
| hardwareId                | string    | Stable identifier of the reporting hardware.                                                                                                   | Required                           |
| source_tank_id            | string    | Tank bound at capture time.                                                                                                                    | Required                           |
| source_owner_uid          | string    | Owner bound at capture time.                                                                                                                   | Required                           |
| source_assignment_at_ms   | number    | Owner and tank assignment recorded when the sensor data was collected.                                                                         | milliseconds since January 1, 1970 |
| captured_at_ms            | number    | Original aggregate completion time.                                                                                                            | milliseconds since January 1, 1970 |
| sensor MIN/MAX/AVG fields | number    | Ten-minute aggregates for temperature, pH, dissolved oxygen, turbidity, and water level; optional operational feed values may also be present. | Only valid sensors included        |
| routing_status            | string    | Result used when the data cannot be safely assigned to a tank.                                                                                 | Optional: quarantined              |
| routing_reason            | string    | Reason the data remains in the temporary receiving area.                                                                                       | Optional                           |